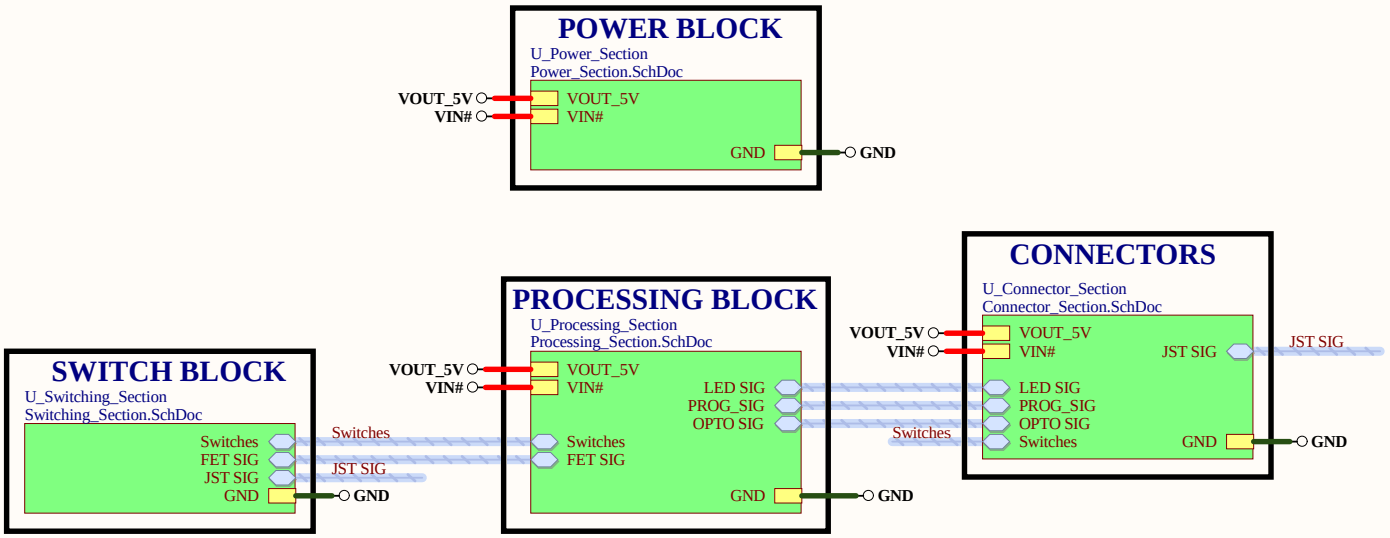
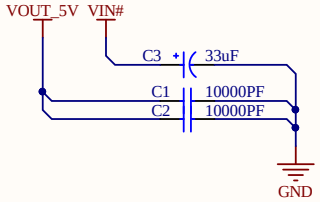
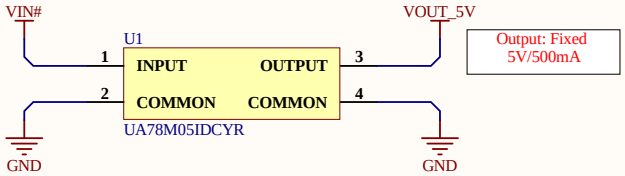

POWER BLOCK
U_Power_Section
Power_Section.SchDoc
VOUT_5V
VIN#
VOUT_5V
VIN#
GND
GND
SWITCH BLOCK
U_Switching_Section
Switching_Section.SchDoc
Switches
FET SIG
JST SIG
GND
GND
PROCESSING BLOCK
U_Processing_Section
Processing_Section.SchDoc
VOUT_5V
VIN#
Switches
FET SIG
LED SIG
PROG_SIG
OPTO SIG
GND
GND
CONNECTORS
U_Connector_Section
Connector_Section.SchDoc
VOUT_5V
VIN#
LED SIG
PROG_SIG
OPTO SIG
Switches
JST SIG
GND
GND
Switches
JST SIG
Switches
JST SIG

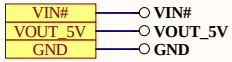
## DECOUPLING CAPACITOR

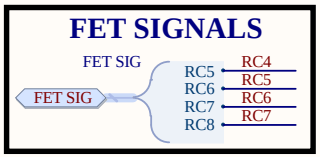


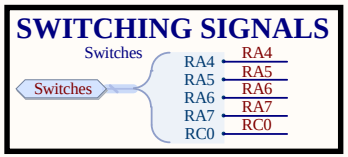

## LINEAR REGULATOR

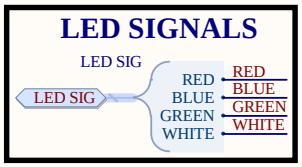


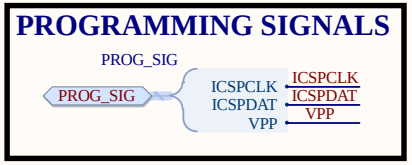

## POWER SIGNALS

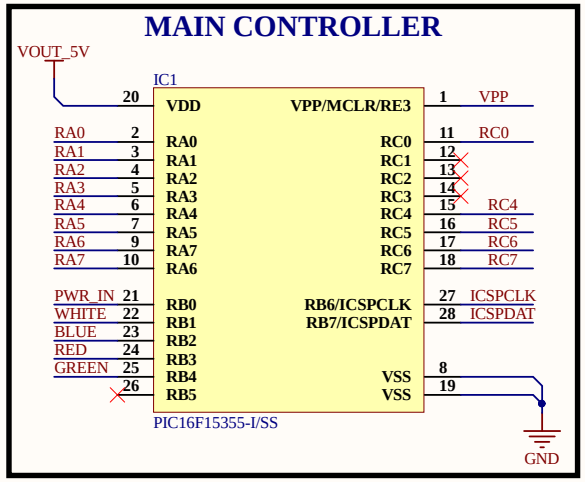


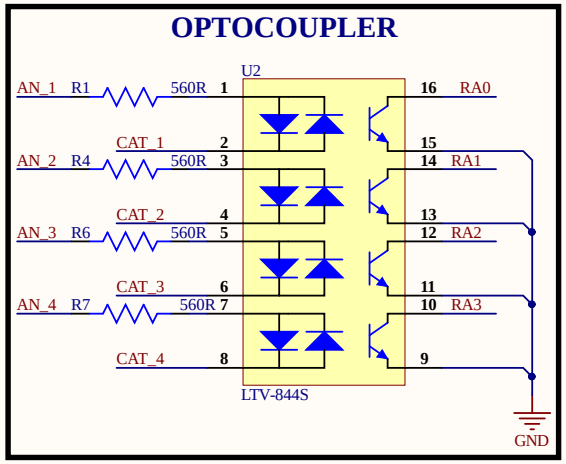

FET SIGNALS
FET SIG
FET SIG
RC5
RC6
RC7
RC8
RC4
RC5
RC6
RC7

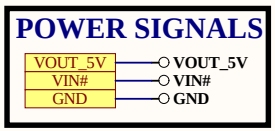

SWITCHING SIGNALS
Switches
Switches
RA4
RA5
RA6
RA7
RC0
RA4
RA5
RA6
RA7
RC0

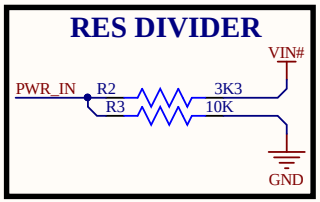

LED SIGNALS
LED SIG
LED SIG
RED
BLUE
GREEN
WHITE
RED
BLUE
GREEN
WHITE

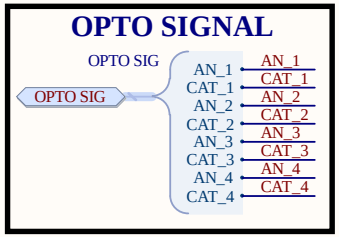

PROGRAMMING SIGNALS
PROG_SIG
PROG_SIG
ICSPCLK
ICSPDAT
VPP
ICSPCLK
ICSPDAT
VPP


MAIN CONTROLLER
VOUT_5V
IC1
20 VDD
VPP/MCLR/RE3 1 VPP
RA0 2 RA0
RA1 3 RA1
RA2 4 RA2
RA3 5 RA3
RA4 6 RA4
RA5 7 RA5
RA6 9 RA7
RA7 10 RA6
RC0 11 RC0
12 RC1
13 RC2
14 RC3
15 RC4
16 RC5
17 RC6
18 RC7
PWR_IN 21 RB0
WHITE 22 RB1
BLUE 23 RB2
RED 24 RB3
GREEN 25 RB4
26 RB5
RB6/ICSPCLK 27 ICSPCLK
RB7/ICSPDAT 28 ICSPDAT
VSS 8
VSS 19
PIC16F15355-I/SS
GND

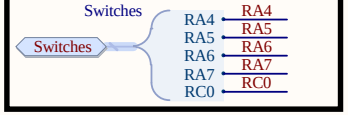

OPTOCOUPLER
U2
AN_1 R1 560R 1
CAT_1 2
AN_2 R4 560R 3
CAT_2 4
AN_3 R6 560R 5
CAT_3 6
AN_4 R7 560R 7
CAT_4 8
16 RA0
15
14 RA1
13
12 RA2
11
10 RA3
9
LTV-844S
GND


POWER SIGNALS
VOUT_5V
VIN#
GND
VOUT_5V
VIN#
GND

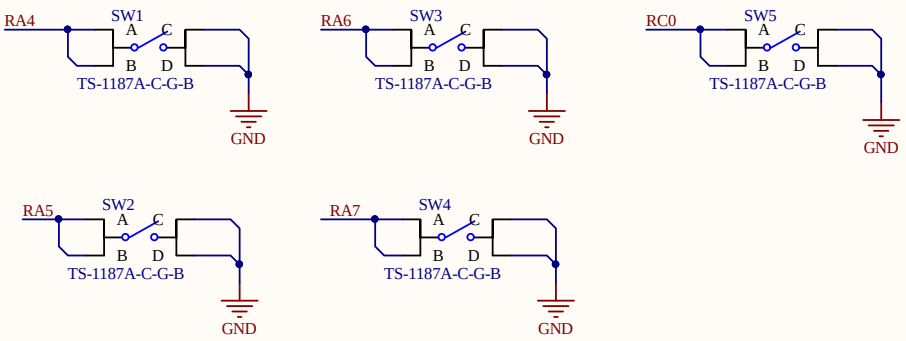

RES DIVIDER
VIN#
PWR_IN
R2
3K3
R3
10K
GND


OPTO SIGNAL
OPTO SIG
OPTO SIG
AN_1
CAT_1
AN_2
CAT_2
AN_3
CAT_3
AN_4
CAT_4
AN_1
CAT_1
AN_2
CAT_2
AN_3
CAT_3
AN_4
CAT_4

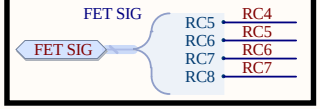
## SWITCHING SIGNALS




## SW HEADERS



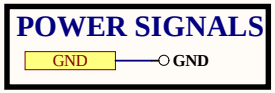

## FET SIGNALS



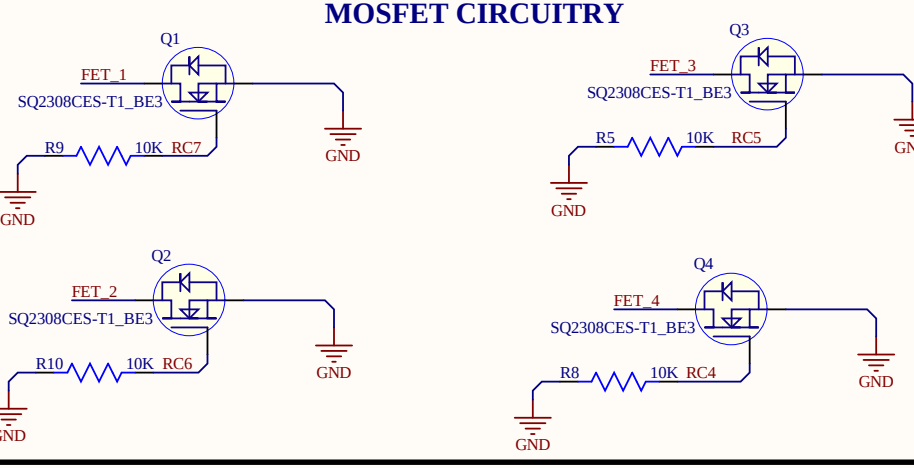

## JST SIGNALS



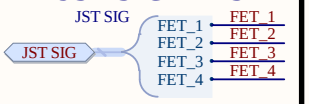

## POWER SIGNALS



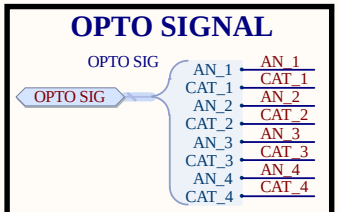

## MOSFET CIRCUITRY



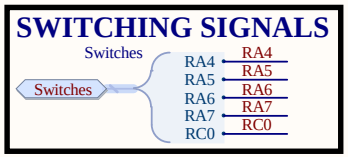
## JST SIGNALS



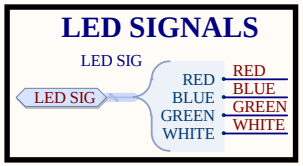

## OPTO SIGNAL



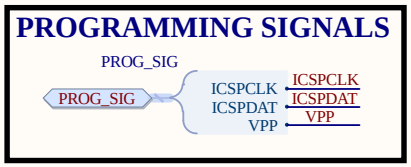

## SWITCHING SIGNALS



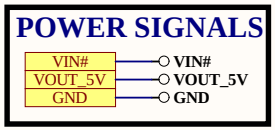

## LED SIGNALS



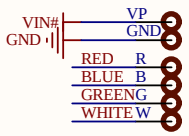

## PROGRAMMING SIGNALS



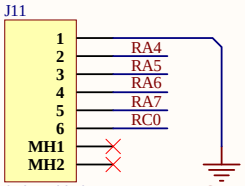

## POWER SIGNALS



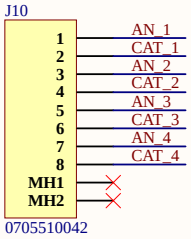

## TEST POINTS



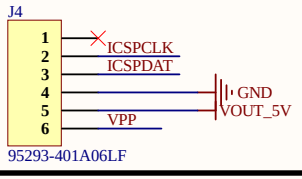

## ON-BOARD BUTTON EXTENSION



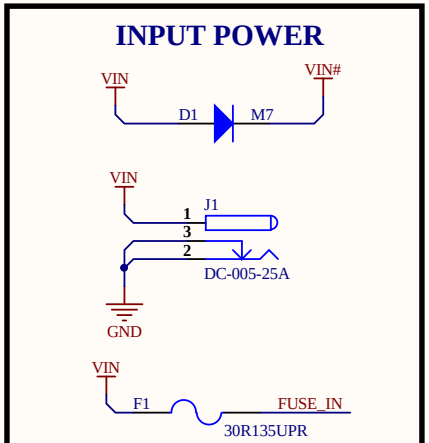

## EXTERNAL SENSE SIGNAL




## PROGRAMMING HEADER




## INPUT POWER




## BOARD TO WIRE HEADERS

